Tooling Validation Report - New Injection Mold - Section 1: Executive Summary & Scope

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 1 (Executive Summary & Scope) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Defines the operational charter, scope boundaries, and executive accountability.

Operational requirement 1.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 1.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 1 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored

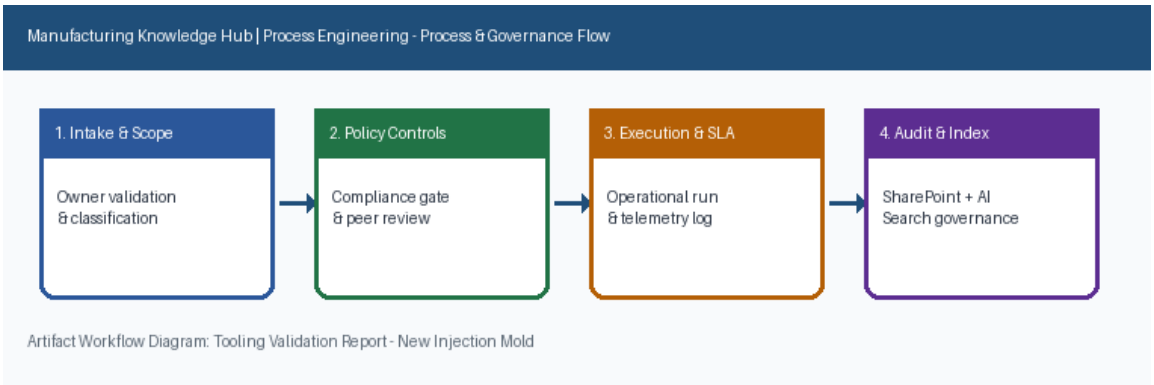
Tooling Validation Report - New Injection Mold - Section 2: Architecture & Process Flow

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 2 (Architecture & Process Flow) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Maps end-to-end process stages, handoff gates, and system-of-record integrations.

Operational requirement 2.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 2.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 2 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-2	Carlos Ibarra	3 business days	Stage 2 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 3: Governance Standards & Controls

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 3 (Governance Standards & Controls) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Establishes mandatory compliance controls, approval thresholds, and audit evidence.

Operational requirement 3.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 3.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 3 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-3	Carlos Ibarra	10 business days	Stage 3 Verified

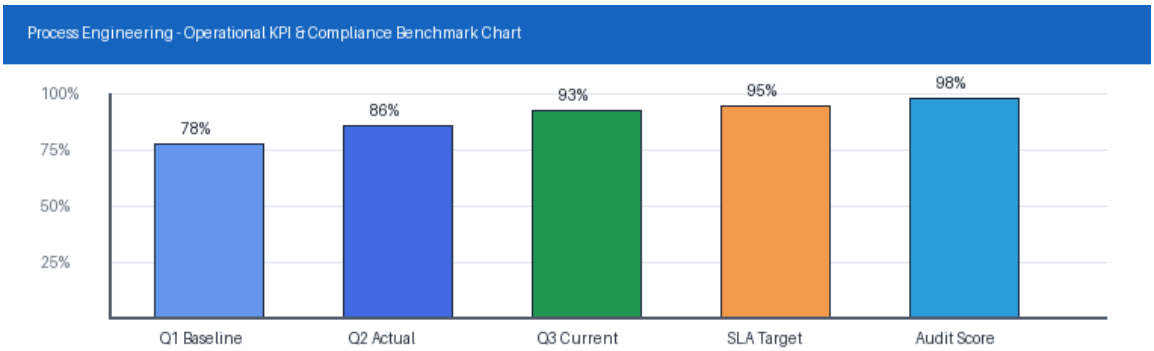
Tooling Validation Report - New Injection Mold - Section 4: Operational KPIs & Benchmarks

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 4 (Operational KPIs & Benchmarks) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Tracks baseline performance, target SLAs, and quarterly benchmark progression.

Operational requirement 4.1: Tool is approved for production release pending preventive maintenance plan sign-off. Secondary control 4.2: First article inspection passed all 14 critical dimensions within tolerance.

- Tool is approved for production release pending preventive maintenance plan sign-off.
- First article inspection passed all 14 critical dimensions within tolerance.
- Section 4 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-4	Carlos Ibarra	5 business days	Stage 4 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 5: Roles, Responsibilities & RACI

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 5 (Roles, Responsibilities & RACI) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Clarifies ownership across accountable executives, operational leads, and reviewers.

Operational requirement 5.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 5.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 5 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-5	Carlos Ibarra	3 business days	Stage 5 Verified

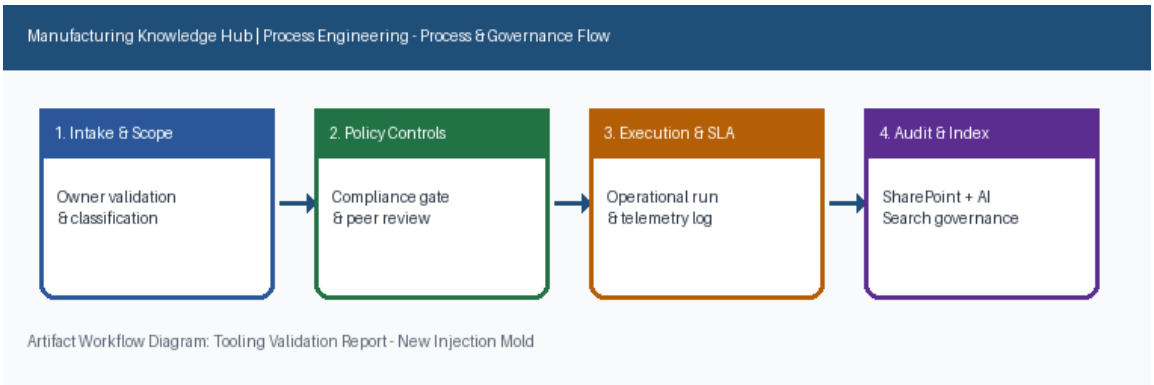
Tooling Validation Report - New Injection Mold - Section 6: Standard Operating Procedures

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 6 (Standard Operating Procedures) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Provides step-by-step execution checklists, intake criteria, and verification gates.

Operational requirement 6.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 6.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 6 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-6	Carlos Ibarra	10 business days	Stage 6 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 7: Risk Register & Mitigations

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 7 (Risk Register & Mitigations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Catalogs operational, compliance, and delivery risks alongside preventive mitigations.

Operational requirement 7.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 7.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 7 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-7	Carlos Ibarra	5 business days	Stage 7 Verified

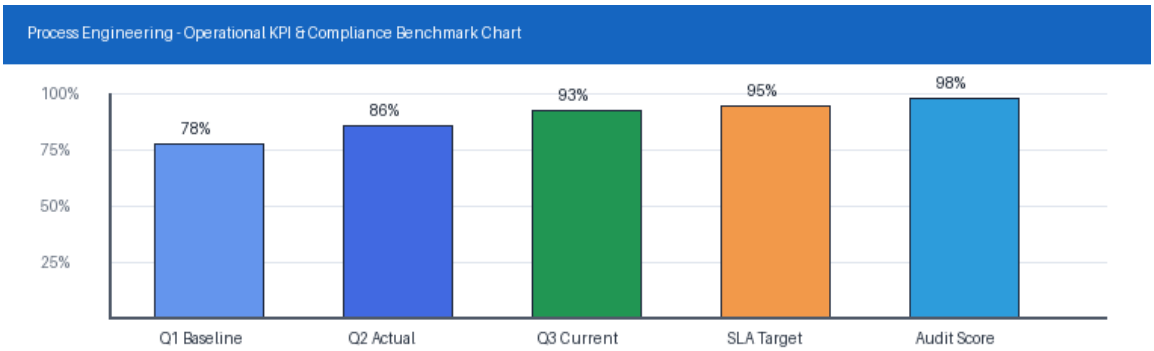
Tooling Validation Report - New Injection Mold - Section 8: Quality Assurance & Audits

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 8 (Quality Assurance & Audits) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Specifies sampling frequencies, inspection tolerances, and nonconformance workflows.

Operational requirement 8.1: Tool is approved for production release pending preventive maintenance plan sign-off. Secondary control 8.2: First article inspection passed all 14 critical dimensions within tolerance.

- Tool is approved for production release pending preventive maintenance plan sign-off.
- First article inspection passed all 14 critical dimensions within tolerance.
- Section 8 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-8	Carlos Ibarra	3 business days	Stage 8 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

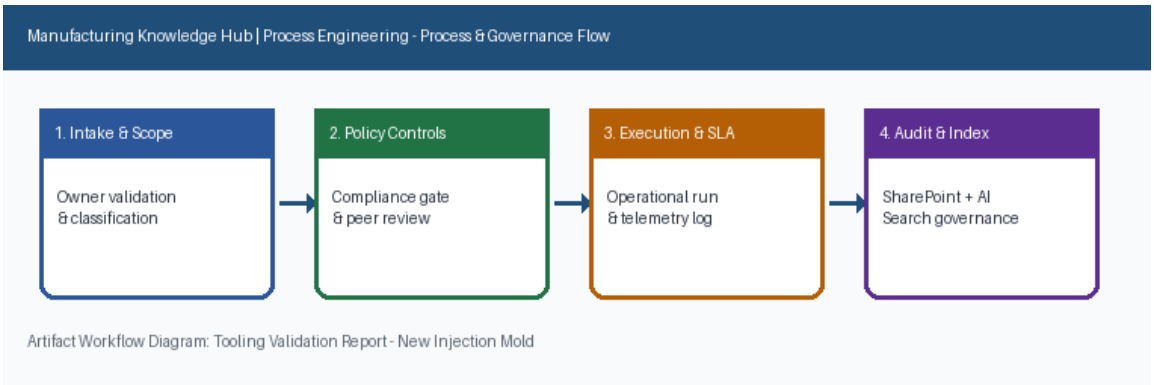
Tooling Validation Report - New Injection Mold - Section 9: Exception Handling & Escalations

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 9 (Exception Handling & Escalations) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Outlines severity tiers, response SLAs, and cross-functional escalation paths.

Operational requirement 9.1: First article inspection passed all 14 critical dimensions within tolerance. Secondary control 9.2: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.

- First article inspection passed all 14 critical dimensions within tolerance.
- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Section 9 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-9	Carlos Ibarra	10 business days	Stage 9 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 10: Cross-Functional Dependencies

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 10 (Cross-Functional Dependencies) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Documents upstream and downstream team dependencies and service-level commitments.

Operational requirement 10.1: Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension. Secondary control 10.2: Mold flow simulation matched actual fill pattern within acceptable variance.

- Short-run capability study produced a Cpk of 1.48 on the primary wall thickness dimension.
- Mold flow simulation matched actual fill pattern within acceptable variance.
- Section 10 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-10	Carlos Ibarra	5 business days	Stage 10 Verified

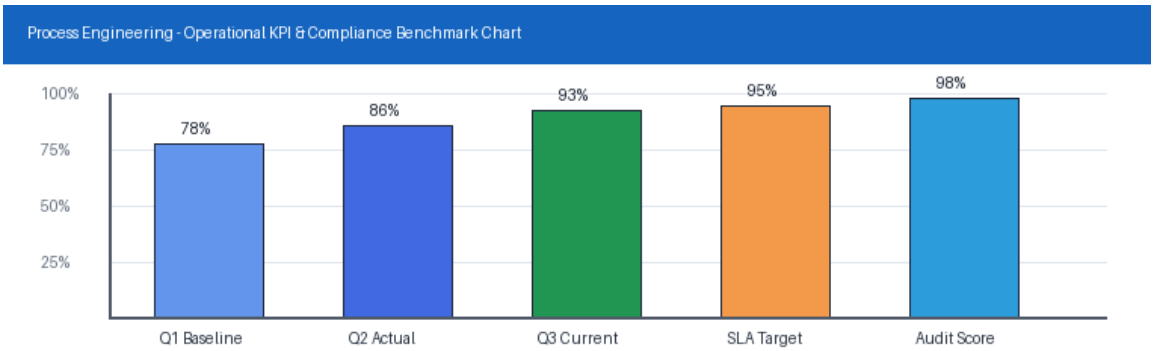
Tooling Validation Report - New Injection Mold - Section 11: Continuous Improvement Roadmap

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 11 (Continuous Improvement Roadmap) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Schedules quarterly optimization milestones, automation targets, and capability reviews.

Operational requirement 11.1: Mold flow simulation matched actual fill pattern within acceptable variance. Secondary control 11.2: Tool is approved for production release pending preventive maintenance plan sign-off.

- Mold flow simulation matched actual fill pattern within acceptable variance.
- Tool is approved for production release pending preventive maintenance plan sign-off.
- Section 11 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-11	Carlos Ibarra	3 business days	Stage 11 Verified



Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online

Tooling Validation Report - New Injection Mold - Section 12: Appendix, Lineage & Sign-off

Validation results for the newly commissioned injection mold tool, covering first article inspection and short-run capability. Section 12 (Appendix, Lineage & Sign-off) applies to Process Engineering within Manufacturing Knowledge Hub and is maintained by Carlos Ibarra. Records document lineage, governance metadata, version history, and owner sign-off.

Operational requirement 12.1: Tool is approved for production release pending preventive maintenance plan sign-off. Secondary control 12.2: First article inspection passed all 14 critical dimensions within tolerance.

- Tool is approved for production release pending preventive maintenance plan sign-off.
- First article inspection passed all 14 critical dimensions within tolerance.
- Section 12 review cadence is tracked by Carlos Ibarra under Process Engineering governance.

Control Area	Owner	Target SLA	Verification Status
Process Engineering	Carlos Ibarra	5 business days	Verified
Tooling Validation Report - Gate 1	Carlos Ibarra	3 business days	Active
Tooling Validation Report - Gate 2	Carlos Ibarra	10 business days	Monitored
Sec-12	Carlos Ibarra	10 business days	Stage 12 Verified